

CODE: Human-Detection PIR Sensor

```
int LED=8;
int PIR=2;
int pirState=LOW;
int val =0;

void setup(){
  pinMode(LED,OUTPUT);
  pinMode(PIR,INPUT);
  Serial.begin(9600);
}

void loop(){
  val=digitalRead(PIR);
  if(val==HIGH){
    delay(1000);
    digitalWrite(LED,HIGH);

    if(pirState == LOW){
      Serial.println("Motion detected!");
      pirState =HIGH;
      delay(1000);
    }
  }
  else{
    digitalWrite(LED,LOW);
    if(pirState == HIGH){
      Serial.println("Motion ended!");
      pirState=LOW;
      delay(1000);
    }
  }
}
```

OUTPUT:

```
Motion detected!
Motion ended!
Motion detected!
Motion ended!
Motion detected!
Motion ended!
Motion detected!
Motion ended!
Motion detected!
Motion ended!
Motion detected!
Motion ended!
```

CODE: Intensity Measurement

```
int LED=12;  
int LDR=A0;  
  
void setup(){  
  Serial.begin(9600);  
  pinMode(LED,OUTPUT);  
  pinMode(LDR,INPUT);  
  
}  
  
void loop(){  
  delay(2000);  
  int LDRStatus = analogRead(LDR);  
  Serial.println(LDRStatus);  
  if(LDRStatus >= 800){  
    digitalWrite(LED,HIGH);  
    Serial.println("LED ON!");  
  }  
  else{  
    digitalWrite(LED,LOW);  
    Serial.println("LED OFF!");  
  }  
}
```

OUTPUT:

413
LED OFF!
413
LED OFF!
413
LED OFF!
883
LED ON!
963
LED ON!
911
LED ON!
911
LED ON!
911
LED ON!
911
LED ON!
911
LED ON!
268
LED OFF!
268
LED OFF!
268
LED OFF!

CODE: Build a Circuit on the Breadboard – LED Display (infinte times)

```
int LED = 8;
void setup() {
  pinMode(LED, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(LED, HIGH);
  Serial.print("LED ON! ");
  delay(1000);
  digitalWrite(LED, LOW);
  Serial.print("LED OFF! ");
  delay(1000);
}
```

OUTPUT:

LED ON!
LED OFF!
LED ON!
LED OFF!
LED ON!
LED OFF!

FOR 5 TIMES BLINKING WE NEED TO GIVE THE CODE IN THE SETUP PART WITH THE FOR LOOP

FOR BLINKING THE BUILT-IN LED WE NEED TO SET THE Digital Pin 13.